

Cell Birth, Cell Death & Tissue Homeostasis - Factoid Sheet

Week 4 - Wednesday - Core Lecture 10 - current faculty deck + official Week 4 objectives

Master chain: growth signal → Rb restriction point → DNA-damage p53 checkpoint → repair arrest or apoptosis → tissue homeostasis.

Rb Restriction Point

Active/hypophosphorylated **Rb binds E2F** → S-phase genes OFF. Growth factor → Cyclin D-CDK4/6 → Rb phosphorylation/inactivation → E2F released → S phase.

Rb is a tumor suppressor. Loss of both functional alleles removes the brake; excess Cyclin D/CDK4/6 can functionally inactivate Rb. CDK4/6 inhibitors exploit this pathway.

Flow Cytometry

Phase	DNA
G1	2N
S	2N → 4N
G2/M	4N
Sub-G1	<2N; fragmented DNA/apoptosis clue

Propidium iodide intercalates into DNA; fluorescence intensity reports DNA content. CDK4/6 inhibition → G1 accumulation. Taxol → G2/M accumulation.

p53 Decision

Lower damage: p53 → p21 → CDK inhibition → growth arrest/repair.

High/persistent damage: p53 → Bax → mitochondrial apoptosis.

Apoptosis - Fast Recall

Intrinsic Pathway

Stress/p53 → Bax/Bak → mitochondrial permeabilization → cytochrome c → **caspase-9** → executioner caspases 3/6/7.

Anti-apoptotic	Pro-apoptotic
Bcl-2, Bcl-XL, Mcl-1	Bax, Bak, Bid, Bim

Extrinsic Pathway

FasL/death ligand → Fas/death receptor + FADD → **caspase-8/10** → executioners. Caspase-8 can cleave Bid → tBid → mitochondrial amplification.

Apoptosis Detection

Marker	Meaning
Annexin V	Externalized phosphatidylserine
PI	Membrane permeability / DNA-content context
Cleaved caspase / lamin	Proteolytic execution
DNA ladder / sub-G1	Fragmented nucleosomal DNA

Apoptosis vs Necrosis

Apoptosis: shrinkage, chromatin condensation, intact membrane until late, PS outside, apoptotic bodies.	Necrosis: swelling, early membrane rupture, leakage, inflammation, no classic DNA ladder.
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Must-Not-Miss

Rb ON = E2F OFF. G1 2N; G2/M 4N. p53-p21 = arrest. p53-Bax = death. Intrinsic = caspase-9. Extrinsic = caspase-8/10. Bcl-2 blocks death; Bax/Bak promote it. Annexin V = PS.